

Engineering Manager (EM) Take-Home Assignment

Congratulations on advancing to this stage of our interview process at Careerflow.ai! We are thrilled to see how you tackle a real-world technical challenge that mirrors the strategic and technical responsibilities of the EM role.

Background

Careerflow.ai is revolutionizing the job search process with AI-driven tools that include an AI resume builder, cover letter writer, and job application tracking board. One of our key features in the pipeline is the **AI Auto-apply** functionality, designed to help users seamlessly apply to jobs that align with their profile, preferences, and qualifications.

This assignment will focus on conceptualizing, planning, and prototyping this feature.

Assignment Overview

Objective

Design a comprehensive strategy and prototype for the **AI Auto-apply** feature, demonstrating your ability to merge strategic vision with hands-on technical expertise.

Deliverables

1. Technical Design Document

- **Feature Overview:** Provide a high-level description of the AI Auto-apply feature, including its purpose, target users, and expected outcomes.
- **System Architecture:** Outline the architecture for the feature, including all the different components.
- **AI Integration:** Detail how AI will be used to analyze user profiles, job descriptions, and application requirements to automate the application process effectively.
- **Scalability:** Include a plan for scaling the feature to handle millions of users.

2. Prototype Implementation

- Develop a lightweight prototype that demonstrates the ability to work across the stack for the AI Auto-apply feature.
- **Frontend Requirements:**

- Use React and TypeScript to build a minimal user interface. Include a single button to trigger the auto-apply feature.
- Display the application status (e.g., success, failure, pending) in real-time.
- Hardcode preferences such as keywords, job filters, and application settings for simplicity.
- **Backend Requirements:**
 - Use Firebase as the database to store user preferences and application history.
 - Implement a lightweight backend service using a framework of your choice (e.g., Express.js, FastAPI) to simulate job scraping, processing, and actually applying.
 - Integrate a basic AI model or API (e.g., OpenAI, Claude) in this solution.
- Focus on showing how front-end and back-end systems interact seamlessly, rather than building a fully functional auto-apply system.

3. Implementation Roadmap and Project Plan

- Provide a comprehensive roadmap and project plan for building and launching the AI Auto-apply feature. This should include:
 - **Team Requirements:** Specify the roles and skills needed (e.g., front-end developer, back-end engineer, AI/ML specialist).
 - **Timeline:** Break the project into phases with estimated timelines for each (e.g., research, design, implementation, testing, deployment), including key milestones.
 - **Phases and Deliverables:** Clearly define deliverables for each phase, such as prototypes, code reviews, and testing reports.
 - **Task Management:** Create a task board (e.g., Linear, Jira, Asana) or a Gantt chart to illustrate project flow and dependencies.
 - **Resource Planning:** Include considerations for tools, APIs, and any other resources required.
 - **Leadership and Execution:** Highlight how you would transition the project into execution mode, focusing on leadership, coordination, and adaptability.

4. Risk Assessment

- Identify potential risks associated with the development and deployment of the AI Auto-apply feature. These might include:
 - **Job Matching Errors:** Risks related to inaccurate matches between user profiles and job descriptions. Mitigation strategies could include implementing robust AI model testing, data validation, and user feedback loops.
 - **API Limitations and Reliability:** Challenges with third-party API rate limits, downtimes, or data consistency issues. Propose strategies such as caching, retry mechanisms, and alternative data sources.

- **Data Security and Compliance:** Risks involving user data breaches or non-compliance with regulations like GDPR. Include mitigation strategies such as encryption, secure authentication methods, and periodic compliance audits.
 - **Scalability Challenges:** Potential performance issues as user volume grows. Propose solutions such as horizontal scaling, database optimization, and performance monitoring tools.
 - **User Experience Risks:** Issues that may arise from unclear workflows or UI design flaws. Mitigation could involve usability testing and iterative design improvements.
- For each risk, outline the probability of occurrence, potential impact, and specific steps to reduce or manage the risk effectively.
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How Long It Should Take

We expect you to complete the task within **24 hours**. If you need additional time or cannot start immediately, please let us know.

Final Deliverables

- **Technical Design Document:** PDF detailing the architecture, AI approach, scalability plan, and compliance measures.
 - **Prototype Code:** Upload to a private repository or send as a zip file.
 - **Project Plan:** A comprehensive project plan, including timelines, team requirements, resource allocation, and task management tools (e.g., Linear, Jira, Gantt chart).
 - **README File:** Include setup instructions, architecture explanation, and testing notes.
 - **Video Walkthrough:** A 5-10 minute video explaining the assignment.
 - **Responsible Use of AI Statement:** Please include a short "Responsible Use of AI" statement with your submission, mentioning how (if at all) you used AI tools (e.g., ChatGPT, Copilot, etc.) — for example, for debugging, idea generation, or code assistance. This helps us ensure transparency and responsible AI use. Note that if you are also using AI to generate code, then you need to also share with us full chat / prompt traces for transparency about how much code you actually wrote vs an AI system.
 - We look for individuals who are great at leveraging AI technology to help them in their work, while still maintaining expertise over the subject matter, and knowing when to trust their own instincts and judgement vs relying fully on an AI assistant.
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Setting Expectations

We understand that this assignment covers multiple sections, and completing all parts is not mandatory. The primary goal is to gain insights into your strengths and areas of expertise.

While each section provides valuable information about different aspects of your skill set, you are encouraged to focus on the parts where you feel most confident and can showcase your abilities effectively.

That said, candidates who complete all sections and deliverables with thoroughness and attention to detail will be regarded highly in the evaluation process.

Remember, we are looking for a candidate that demonstrates strategic thinking, technical proficiency, and alignment with Careerflow.ai's mission. Focus on delivering a structured and thoughtful approach that highlights your ability to lead technical projects and solve real-world problems.

The Purpose of This Assignment

This task provides insights into your ability to:

- Translate high-level business goals into technical strategies.
- Design and implement AI-driven solutions.
- Balance hands-on technical work with long-term vision.
- Address challenges related to scalability, compliance, and security.
- Lead a team effectively from planning to execution.

Additionally, this assignment offers you an opportunity to:

- Gain a deeper understanding of the level of work expected in this role.
- Familiarize yourself with the types of challenges and expectations at Careerflow.ai.
- Learn about our focus on innovation, scalability, and impact through the design and execution of meaningful features.

For any queries, concerns, or clarifications, please reach out to puneet@careerflow.ai
